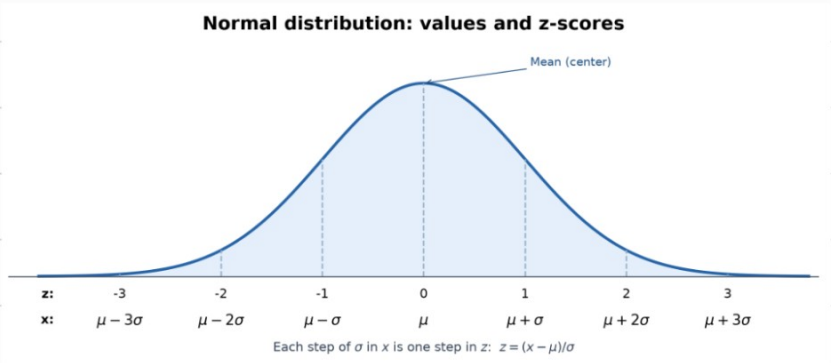
8.1 Standard Normal Distribution

1. The Normal Distribution

- A **Bell-Shaped**, symmetric curve written $X \sim N(\mu, \sigma)$; the first number is the **mean**, the second the **Standard Deviation**.
- The mean sits at the center. Each standard deviation away from it is one unit of **Z-Score**.



2. Key Formulas

Purpose	Formula
Z-Score of a Value x	$z = \frac{x-\mu}{\sigma}$
Value x From a Z-Score	$x = \mu + z\sigma$
PDF (for reference)	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$

You don't need the PDF to find probabilities in an intro course. Use z-tables (or calculus, see Lesson 8.2)

- **Positive** $z \rightarrow$ the value **above** the mean, **Negative** $z \rightarrow$ **below** the mean.

3. The Empirical Rule

Within	Percent of the Data
$\mu \pm 1\sigma$	68%
$\mu \pm 2\sigma$	95%
$\mu \pm 3\sigma$	99.7%

- $68/2 = 34\%$
- $(95 - 68)/2 = 13.5\%$
- $(99.7 - 95)/2 = 2.35\%$
- $(100 - 99.7)/2 = 0.15\%$

The total area under the curve is 1 (100%)

4. Example 1- $X \sim N(50, 10)$

(a) $\mu = 50$, $\sigma = 10$

(b) Which x has $z = 1.4$?

$X = 50 + 1.4(10) = 64$

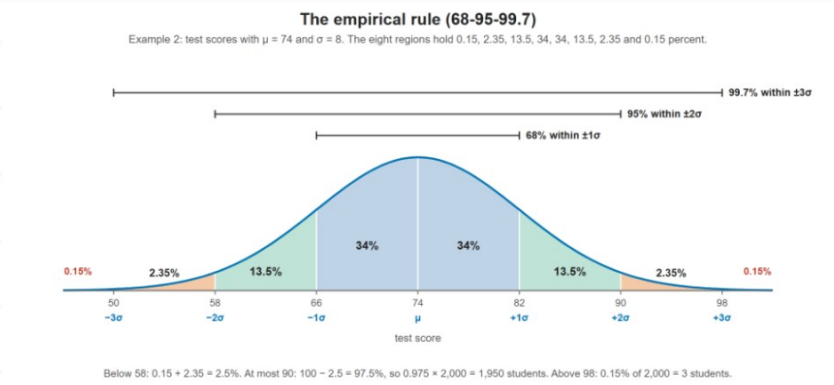
(c) What is the z-score of $x = 30$?

$z = \frac{30-50}{10} = -2$

(d) Negative z-score (like -2 for 30) are left of the mean. Positive ones (like 1.4 for 64) are right of it.

4. Example 2- Empirical Rule: Test Scores, $\mu = 74, \sigma = 8$, 2000 Students

Mark the axis: 50, 58, 66, 74, 82, 90, 98 (from -3σ to $+3\sigma$)



Question	Regions	Answer
(a) % below 58 ($< \mu - 2\sigma$)	0.15 + 2.35	2.5%
(b) $P(66 < X < 82)$ ($\mu \pm 1\sigma$)	34 + 34	68%
(c) How Many scores at most 90?	Everything below $+2\sigma$ = 100 − (2.35 + 0.15) = 97.5%	0.975 × 2000 = 1950 Students
(d) % at least 66	34 + 34 + 13.5 + 2.35 + 0.15	84%
(e) How Many scores more than 98?	0.15%	0.0015 × 2000 = 3 Students

5. Using Z-tables

When z is not a whole number, such as $z = 1.56$, the empirical rule can't be used. Use a standard normal z table instead.

To Find	Use
$P(X < x)$	The table value directly
$P(X > x)$	1 − table value
$P(a < X < b)$	Table(z_b) − Table(z_a)

A z-table gives the area to the LEFT of z

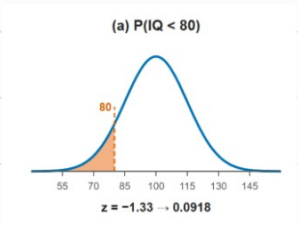
- Find the row for the first (1.5) and the column for the second decimal (0.06): area = 0.94062
- For $z = -0.43$: row -0.4 column 0.03 area = 0.3336 (33.36%)

6. Example 3 - IQ Scores $\mu = 100$, $\sigma = 15$

(a) $P(IQ < 80)$

$$Z = \frac{x - \mu}{\sigma} = \frac{80 - 100}{15} = -1.33 \rightarrow P(Z < -1.33) = 0.09176 = 9.176\%$$

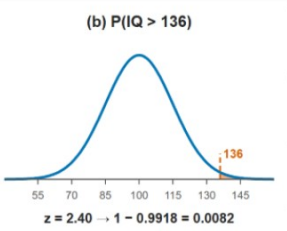
↑ area from Z table



(b) $P(IQ > 136)$

$$Z = \frac{x - \mu}{\sigma} = \frac{136 - 100}{15} = 2.40 \rightarrow P(Z < 2.40) = 0.99180 \leftarrow \text{area from Z table}$$

$$\text{thus } P(X > 136) = 1 - P(Z < 2.40) = 0.0082 = 0.82\%$$

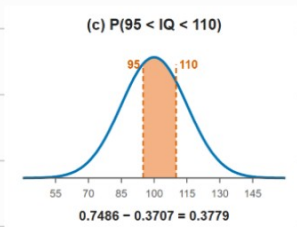


(c) $P(95 < IQ < 110)$

$$Z_{110} = \frac{110 - \mu}{\sigma} = \frac{110 - 100}{15} = 0.67 \rightarrow P(Z < 110) = 0.74857$$

$$Z_{95} = \frac{95 - \mu}{\sigma} = \frac{95 - 100}{15} = \frac{-1}{3} = -0.33 \rightarrow P(Z < 95) = 0.3707$$

$$\text{then } P(95 < IQ < 110) = P(Z < 110) - P(Z < 95) = 37.8\%$$

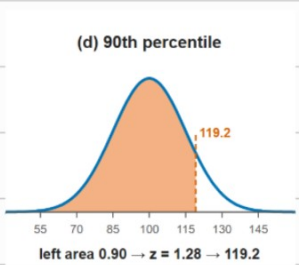


(d) IQ at the 90th Percentile (work backwards from the area)

90th percentile → left Area $P(X < P_{90}) = 0.90$

→ from the table $P = \text{Area} = 0.90 \rightarrow Z = 1.28$

$$Z = \frac{x - \mu}{\sigma} \rightarrow X = \mu + Z \times \sigma = 100 + 1.28 \times 15 = 119.2$$



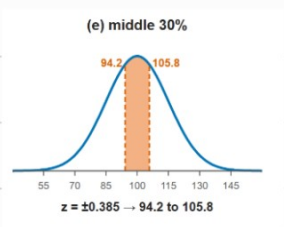
(e) Between which two IQs are the middle 30%?

The middle 30% leaves 15% on Each Side of the mean, so it runs from the 35th to 65th percentile.

- For an area of 0.65 → $Z = 0.385$
- By Symmetry, the 35th percentile is $Z \approx -0.385$

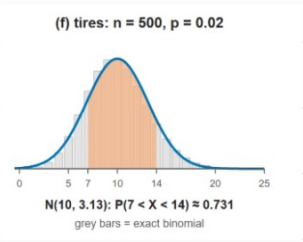
So $X_2 = 100 + 0.385(15) = 105.8$, $X_1 = 100 - 0.385(15) \approx 94.2$

The middle 30% of IQs lie between about 94.2 and 105.8



7. Example 4 - Normal approximation to binomial (defective tires)

2% of tires are defective, and a sample of $n = 500$ is checked. So $p = 0.02$, $q = 0.98$



(a) Mean and Standard Deviation

$$\mu = np = 500(0.02) = 10 \qquad \sigma = \sqrt{npq} = \sqrt{500(0.02)(0.98)} = \sqrt{9.8} \approx 3.13$$

(b) $P(X < 8)$

$$z = \frac{x - \mu}{\sigma} = \frac{8 - 10}{3.13} \approx -0.64 \rightarrow P(z < -0.64) = 0.2611 = 26.11\%$$

(c) $P(X > 15)$

$$z = \frac{x - \mu}{\sigma} = \frac{15 - 10}{3.13} = 1.60 \rightarrow P(z < 1.60) = 0.9452 \rightarrow P(z > 1.60) = 1 - 0.9452 = 5.48\%$$

(d) $P(7 < x < 14)$

$$z_{14} = \frac{14 - 10}{3.13} = 1.28 \rightarrow P(z < 1.28) = 0.89973$$

$$z_7 = \frac{7 - 10}{3.13} = -0.96 \rightarrow P(z < -0.96) = 0.16853$$

$$P(z_7 < z < z_{14}) = P(-0.96 < z < 1.28) = 73.1\%$$

5. Key Takeaways

- $z = \frac{x - \mu}{\sigma}$ and $x = \mu + z\sigma$
- Empirical rule: 68/95/99.7 which split into 34/13.5/2.35/0.15 per region
- A z-table gives left area. Right area = 1 − left. Between = difference of two left areas
- For percentiles, look up the area in the table to find z, then convert to x.

8.2 Standard Normal Distribution with Calculus

1. More Precise Empirical-rule Percentages

The bell curve from $-\infty$ to ∞ with total area 1 (100%). Each half is 50%.

Within	Percent
$\mu \pm 1\sigma$	68.27%
$\mu \pm 2\sigma$	95.45%
$\mu \pm 3\sigma$	99.73%

2. The Exact Method: Integrate the PDF

$$P(a<X<b) = \int_a^b \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

There is no simple antiderivative, so evaluate it with graphing calculator or an online tool such as a definite-integral calculator (e.g. Wolfram|Alpha)

Typing tips: Put the whole exponent in parentheses and include negative sign For $\mu = 75$, $\sigma = 7$

$$e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

For "greater than" questions, the upper limit is $+\infty$, Not the maximum score. Integrating up to 100 underestimates the answer.

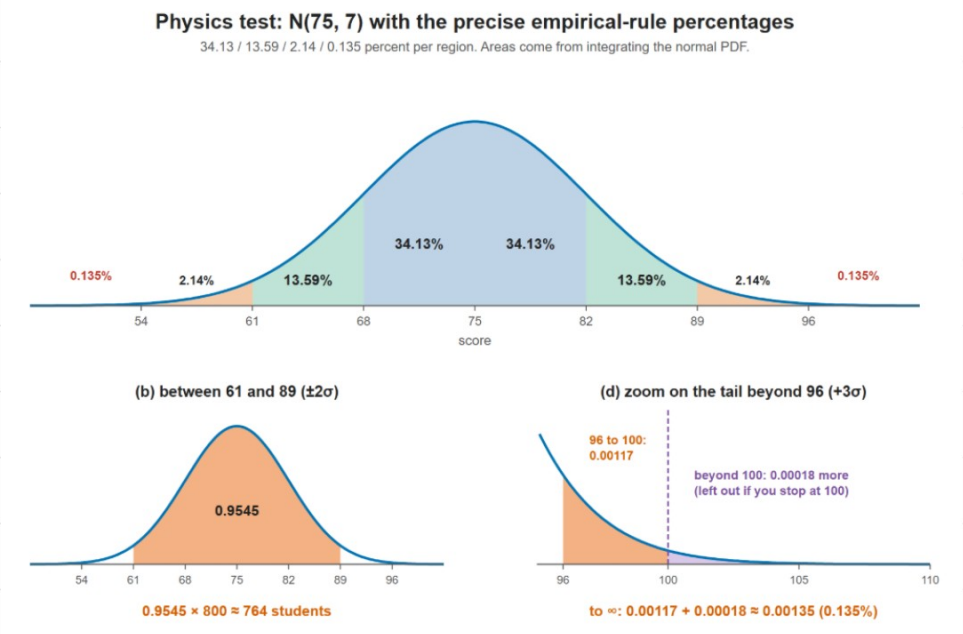
Use very large number (1000; 100000), or $^{\infty}$ if the tool accepts it, and check that the result has stoped changing

3. Example - Physics Test Score

800 Students; Scores are normal with $\mu = 75$, $\sigma = 7$

Axis mark: 54, 61, 68, 75, 82, 89, 96

-3σ -2σ $-\sigma$ μ $+\sigma$ $+2\sigma$ $+3\sigma$



(a) What Percentage Scores between 68 and 82? ($\pm 1\sigma$)

- Empirical rule: $34.134\% + 34.134\% = 68.27\%$
- Integral from 68 to 82: $0.682689 = 68.27\%$

(b) About how many student Scored between 61 and 89? ($\pm 2\sigma$)

- Empirical rule: $13.59 + 34.13 + 34.13 + 13.59 \approx 95.45\%$
- Integral from 61 to 89: 0.954500
- Number of Student: $0.9545 \times 800 \approx 763.6 \rightarrow$ About 764 Students

(c) P(Score between 54 and 75) (-3σ to the mean)

- Empirical rule: $2.14 + 13.59 + 34.13 = 49.87\%$
- Integral from 54 to 75: $0.498650 \rightarrow 49.87\%$

(d) About how many student Scored 96 or more? (Beyond $+3\sigma$)

- Empirical rule: 0.135%
- Integral from 96 to 100: 0.00117. This is too small because the curve continues past 100
- Integral from 96 to 1000 (effectively ∞): $0.0013499 = 0.135\%$
- Number of students: $0.0013499 \times 800 \approx 1.08 \rightarrow$ about 1 student

4. Key Takeaways

- More Precise empirical-rule values: 68.27/95.45/99.73 which split into 34.13/13.59/2.14/0.135 per region.
- The exact probability is the integral of the normal PDF between the limits. Use technology.
- For tails, Integrate out to Infinity, using very large number if needed.
- To get a Count, multiply the probability by the group size and round to a whole number.

8.3 Law of Large Numbers

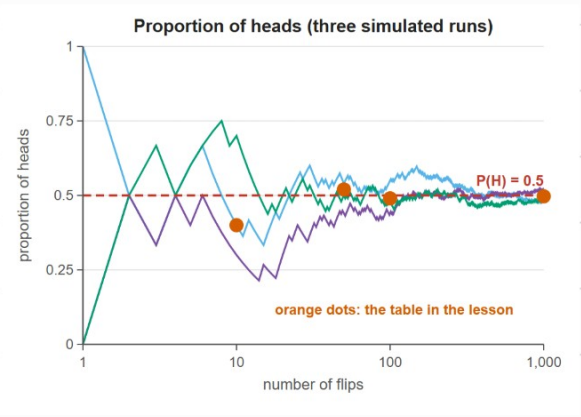
1. Statement

If an experiment is repeated many times, the observed average (or observed proportion) gets closer and closer to the true, theoretical value as the number of trials increases.

2. Example 1 - Flipping Coin

The theoretical probability of heads is $P(H) = \frac{1}{2} = 0.5$. A few trials can stray far from 0.5, but many trials land close to it:

Number of Flips	Heads Observed (example)	Observed Proportion
10	4	0.40
50	26	0.52
100	49	0.49
1000	497	0.497



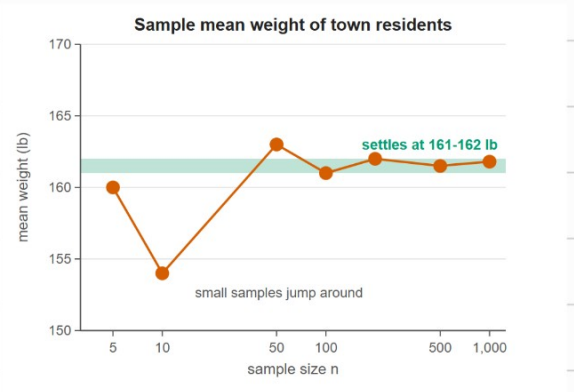
As the number of flips grows, the observed proportion Approaches 0.5.

3. Example 2 - Estimating an Average Weight.

A town has 20000 resident. Instead of Weighting everyone, take Sample of Increasing Size:

- Small sample jump around, because one unusually heavy (230 lb) or light (110 lb) person has a big effect.
- Large Sample settle down. Here they converge to about 161-162 lb, which estimate population mean.

Sample Size n	Sample mean Weight
5	160 lb
10	154 lb
50	163 lb
100	161 lb
200	162 lb
500	161.5 lb
1000	161.8 lb



4. Key Takeaways

- More trials or a bigger sample → The observed average get closer to the true Average.
- This is why large samples give reliables estimate of population parameter
- The law say nothing about any single trial. Even after many heads in a row, the next flip is still 50/50.

8.4 The Central Limit Theorem

1. The Idea

1. Take many samples of size n from a population.
2. Compute the mean $\bar{x}$ of each sample.
3. Plot all those sample mean in histogram.

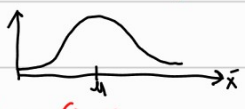
That histogram is the Sampling distribution of the sample mean, and it is approximately Normal.

Population Shape	Sampling Distribution of $\bar{x}$
- Normal	- Normal for any n
- Anything else (Uniform, exponential, Skewed, Irregular)	- Approximately normal when n is large enough. $n \geq 30$ is the usual rule of thumb

Example: in a town of 100,000. We repeated samples of 30 people and record each sample' mean age (41.8, 39.6, 40.5, ...) Those means form a bell curve even if individual ages do not.

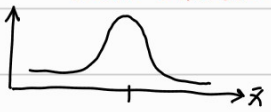
- $n=30$, $\bar{x}_1 = 41.8$, $\bar{x}_2 = 39.6$, $\bar{x}_3 = 40.5$, ... The mean will take shape as normal distribution.

- Same Shape
  - Population distribution

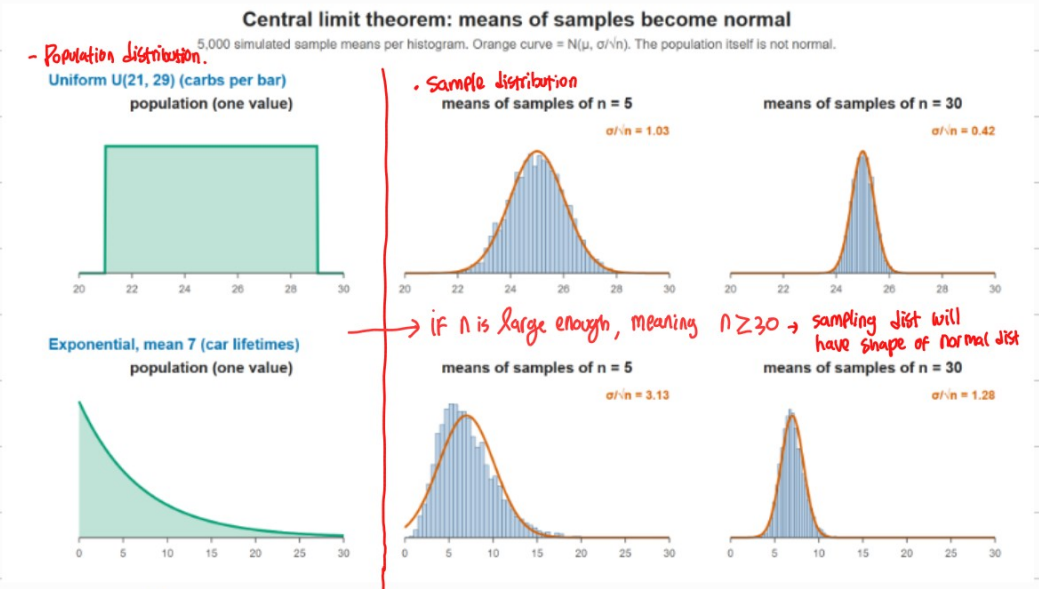


for any n value

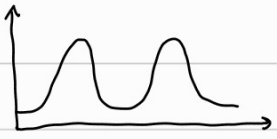
  - Sample distribution



- Different Shape

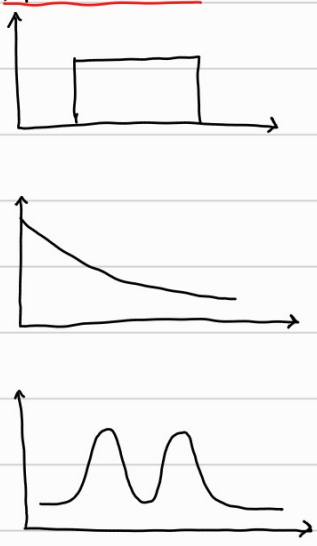


- Bimodal distribution



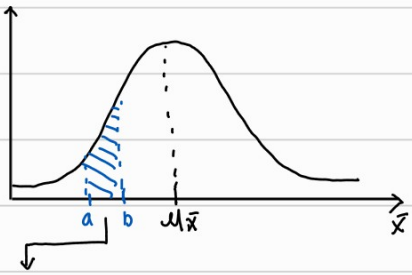
Neither population is normal, yet the sample means pile up into a bell curve. With $n=30$ they match the normal curve $N(\mu, \sigma/\sqrt{n})$ closely, even for the skewed exponential population, where $n=5$ is still visibly lopsided.

Population dist.



If $n \geq 30$

Sampling Dist



We can use Z-table to calculate the probability that sample mean is between a and b.

2. Symbols

Symbol	Meaning
μ	Population Mean
σ	Population Standard Deviation
$\bar{x}$	Mean of One Sample
s	Standard Deviation of One Sample
$\mu_{\bar{x}}$	Mean of the Sampling distribution: $\mu_{\bar{x}} = \mu$
$\sigma_{\bar{x}}$	SD of the sampling distribution, the Standard error: $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$
n	Sample Size

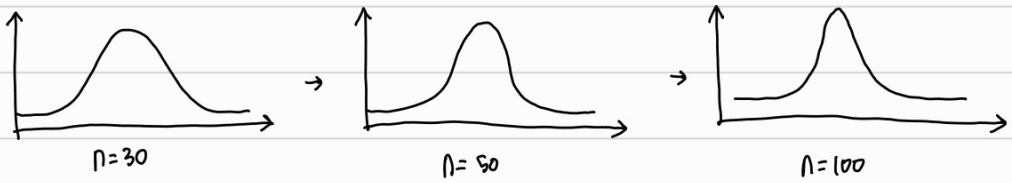
Law of large number ink: As n grows, $\bar{x}$ gets closer to μ. For example, mean ages from samples of 10, 50, 100 and 1000 people might be 32, 38.5, 35.3 and 36.1 approaching the true 36.

3. Larger n → Narrower Curve

$\sigma_{\bar{x}} = \sigma/\sqrt{n}$

n is in the denominator, so a larger sample gives a smaller standard error. The sampling distribution become taller and narrower

$n = 50$ is narrower than $n = 30$, which is narrower than $n = 10$.



$n \uparrow$ $\quad \bar{x} \rightarrow \mu$

$n \uparrow$ $\quad \mu_{\bar{x}} \rightarrow \mu$

$n \geq 30$ $\quad \mu_{\bar{x}} = \mu$

4. Z-Scores: Individuals vs Means vs Sums

We know that $\sigma_{\bar{x}} = \sigma/\sqrt{n}$ so

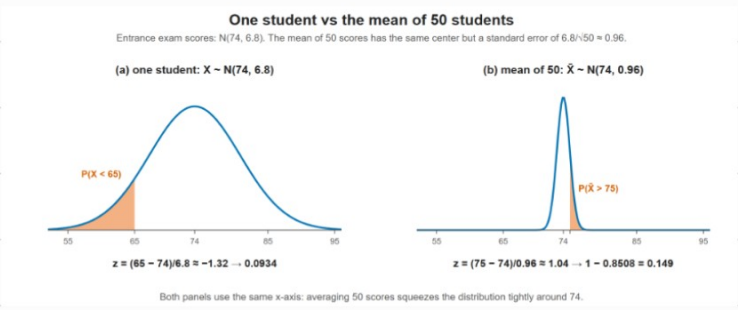
$$Z = \frac{X-\mu}{\sigma} = \frac{X-\mu}{\sigma/\sqrt{n}}$$

Question About...	Distribution	Z-Score
One value x	$X \sim N(\mu, \sigma)$	$Z = \frac{X-\mu}{\sigma}$
The mean of n values	$\bar{X} \sim N(\mu, \frac{\sigma}{\sqrt{n}})$	$Z = \frac{\bar{x}-\mu}{\sigma/\sqrt{n}}$
The Sum of n values	$\sum X \sim N(n\mu, \sqrt{n}\sigma)$	$Z = \frac{\sum x - n\mu}{\sqrt{n}\sigma}$

Reminder from earlier lessons

- Uniform $U(a,b)$: $f(x) = \frac{1}{b-a}$, $\mu = \frac{a+b}{2}$, $\sigma = \frac{b-a}{\sqrt{12}}$, Probability = base x height
- Exponential $f(x) = \lambda e^{-\lambda x}$, $\lambda = 1/\mu$, $\sigma = \mu$, $P(X<x) = 1 - e^{-\lambda x}$

5. Example 1 - Entrance exam Score: Normal, $\mu = 74$, $\sigma = 6.8$.



(a) One student score below 65 ($n=1$, so use the population distribution)

$$Z = \frac{X-\mu}{\sigma} = \frac{65-74}{6.8} = -1.32 \rightarrow P(Z<-1.32) = 0.09342 = 9.34\%$$

(b) The mean of 50 students is above 75

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{6.8}{\sqrt{50}} = 0.9617, \quad Z_{\bar{x}} = \frac{X-\mu}{\sigma_{\bar{x}}} = \frac{75-74}{0.9617} = 1.04$$

$$P(\bar{X}>75) = P(Z_{\bar{x}}>1.04) = 1 - P(Z_{\bar{x}}<1.04) = 1 - 0.85083 = 0.14917 = 14.9\%$$

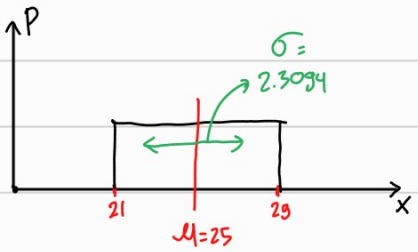
(c) Distribution of the mean of 50 scores

$\mu_{\bar{x}} \approx \mu$ for big number $\quad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = 0.9617$

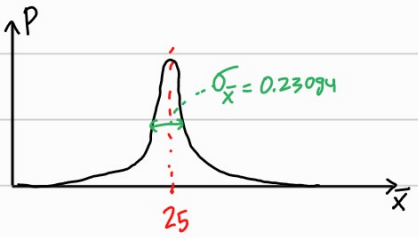
Thus $\bar{X} \sim N(\mu, \sigma_{\bar{x}}) \sim N(74, 0.9617)$

6. Example 2 - Carbs in a Snackbar: Uniform from 21g to 29g, $n = 100$

- Population Distribution



- Sample Mean Distribution



(a) One bar

$X \sim U(21, 29)$, $\mu = \frac{21+29}{2} = 25$ $\sigma = \frac{b-a}{\sqrt{12}} = \frac{29-21}{\sqrt{12}} = 2.309g$

(b) Mean of 100 bars

By the central limit theorem (CLT) $n \geq 30$ the probability distribution function (PDF) is normal although one bar is uniform:

$\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}} = \sigma/\sqrt{n} = 2.309g/\sqrt{100} = 0.2309g$

$\bar{X} \sim N(25, 0.2309g)$

(c) One bar has 24-26 g (uniform, so base × height)

$P(24 < X < 26) = \text{base} \times \text{height} = (26-24) \times f(x)$

$= (26-24) \times \frac{1}{(29-21)} = 2 \times \frac{1}{8} = 0.25$

(d) The mean of 100 bars is between 24.9 and 25.1 g (normal)

The mean of n values: $\bar{X} \sim N(\mu, \sigma/\sqrt{n})$ $Z = \frac{X-\mu}{\sigma/\sqrt{n}}$

$Z = \frac{\pm 0.1}{\sigma_{\bar{x}}} = \frac{\pm 0.1}{0.2309g} = \pm 0.43$

$P(24.9 < \bar{x} < 25.1) = P(\bar{x} < 25.1) - P(\bar{x} < 24.9)$

$= 0.6640 - 0.33360 = 0.3328$

(e) Distribution of the total carbs in 100 bars (normal)

The sum of n values: $\sum X \sim N(n\mu, \sqrt{n}\sigma)$ $Z = \frac{nX - n\mu}{\sqrt{n}\sigma}$

$\sum X \sim N(100 \cdot 25;\ 10 \cdot 2,309g) = N(2500, 23.09g)$

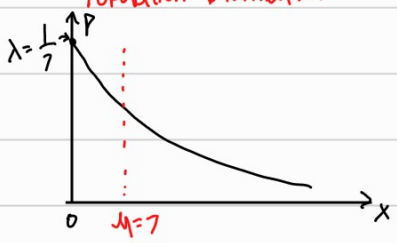
(f) The total is more than 2540 g

$Z = \frac{X-\mu}{\sigma} = \frac{2540-2500}{23.09g} \approx 1.73$

$P(\sum \bar{X} > 2540) = 1 - P(\sum \bar{X} < 2540) = 1 - 0.95818 = 0.04182 \approx 4.2\%$

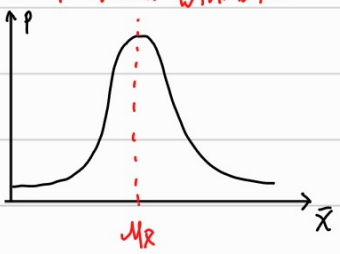
7. Example 3 - Car lifetimes: Exponential with $\mu = 7$ years, $n = 40$ Cars

- Population Distribution



for large $n \geq 30$ →

- Sample Mean Distribution



(a) Rate and SD (one car)

Since it is one car then it will be population distribution

We know - rate $\lambda = \frac{1}{\mu} = \frac{1}{7}$

- Standard Deviation $\sigma = \mu = 7$ years

(b) Distribution of the mean lifetime of 40 cars (CLT: normal)

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{7}{\sqrt{40}} \approx 1.1068 \qquad \bar{X} \sim N(7, 1.1068)$$

(c) P(Sample mean < 6.5 years)

$$Z_{\bar{x}} = \frac{x - \mu_{\bar{x}}}{\sigma_{\bar{x}}} = \frac{x - \mu}{\sigma_{\bar{x}}} = \frac{6.5 - 7}{1.1068} = -0.45$$

$$P(\bar{X} < 6.5) = 0.32636 \approx 32.6\%$$

(d) IQR of the Sample mean

- Q_1:
  - $P_{left} = 0.25 \rightarrow Z_{\bar{x}} = -0.675$

    We know that $Z_{\bar{x}} = \frac{x - \mu_{\bar{x}}}{\sigma_{\bar{x}}} \rightarrow \bar{x} = Z_{\bar{x}} \sigma_{\bar{x}} + \mu_{\bar{x}}$

    $\bar{X}_{Q_1} = -0.675 \times 1.1608 + 7 \approx 6.253$
- Q_3:
  - $P_{left} = 0.75 \rightarrow Z_{\bar{x}} = +0.675$

    $\bar{X}_{Q_3} = 0.675 \times 1.1608 + 7 \approx 7.747$
- $IQR = Q_3 - Q_1 = 7.747 - 6.253 = 1.494$ years

8. Key Takeaways

- Sample means are approximately normal for $n \geq 30$, whatever the population's shape (exactly normal if the population is normal)
- $\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}} = \sigma/\sqrt{n}$, So a bigger n gives a smaller spread
- Decide whether the question is about one value (use σ and the population's own shape), a mean (use $\sigma/\sqrt{n}$ normal) or a Sum (use $n\mu$ and $\sqrt{n}\sigma$, normal)

8.5 The Standard Error of the Mean

1. Formula

The Standard error of the mean (SE) is the standard deviation of sampling distribution $\bar{x}$. It measure how much sample means vary from sample to sample.

You Know	Standard Error
The Sample SD s	$SE = s/\sqrt{n}$
The Population SD σ	$\sigma_{\bar{x}} = \sigma/\sqrt{n}$

2. Example 1 - From sample data

The ages of 5 randomly selected college student are 18, 21, 19, 20, 26

(a) Sample Standard Deviation

Sample Mean $\bar{x} = \frac{\Sigma x}{n} = \frac{104}{5} = 20.8$

Sample Standard Deviation $s = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n-1}} = \sqrt{\frac{38.8}{5-1}} \approx 3.114$

(b) Standard Error

$SE = \frac{s}{\sqrt{n}} \approx \frac{3.114}{\sqrt{5}} \approx 1.393$

3. Example 2 - From the population SD

At a University, Student ages have mean 20.5 and Population SD $\sigma = 0.8$.

(a) Samples of $n = 25$

$$\sigma_{\bar{x}} = \frac{0.8}{\sqrt{25}} = \frac{0.8}{5} = 0.16$$

(b) Samples of $n = 100$

$$\sigma_{\bar{x}} = \frac{0.8}{\sqrt{100}} = \frac{0.8}{10} = 0.08$$

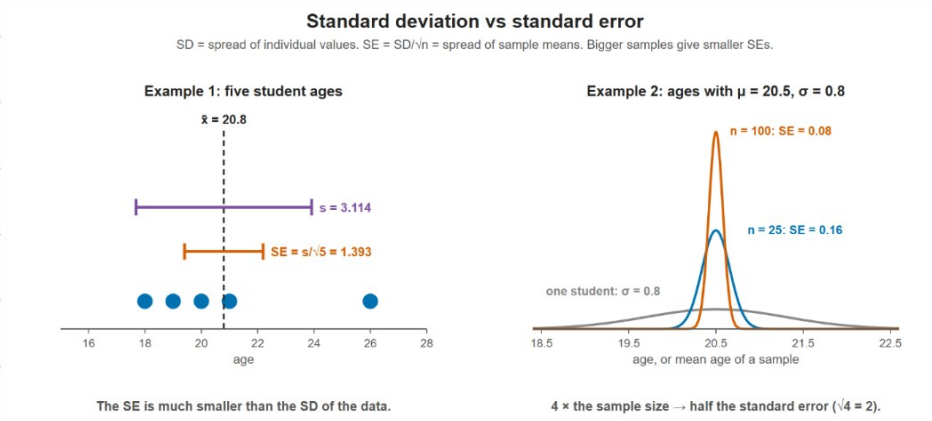
Quadrupling n halves the error, because $\sqrt{4} = 2$

4. What it looks like

Plot the means of many samples to get the Sampling distribution of the mean, a bell curve centered at μ

- With $n = 25$ it is wider and shorter
- With $n = 50$ (or 100) it is narrower and taller

A smaller SE means the sample means cluster more tightly around μ, so any one sample mean is more accurate estimate.



Left: the SD describe how spread out the five ages are, while SE describe how much the mean would vary.

Right: going from $n: 25$ to $n: 100$ halves the SE, so the curve of sample means get twice as narrow.

5. Key Takeaways

- $SE = \frac{SD}{\sqrt{n}}$ Use s from a sample or σ from the population
- The SD describes spread among Individual value. The SE describes spread among sample means.
- $SE \propto \frac{1}{\sqrt{n}}$: a large sample gives smaller SE and a more precise estimate.